

Database Projects Portfolio

This repository is a professional showcase of database-related projects, automation scripts, and sample queries demonstrating my skills in database administration, automation, and maintenance.

Repository Structure

```
database-projects/
├── README.md
├── scripts/
│   ├── backup/
│   │   └── mysql_backup.sh
│   ├── maintenance/
│   │   └── clean_old_records.sql
│   └── automation/
│       └── user_report_procedure.sql
├── projects/
│   ├── migration_project/
│   │   ├── migration_notes.md
│   │   └── migration_script.sql
│   └── analytics_demo/
│       └── sample_queries.sql
└── docs/
    └── database_architecture_diagram.png
```

Project Highlights

1. Production Database Migration

- Migrated MySQL 5.7 to 8.0
- Managed 45GB production database
- Automated backup and validation scripts
- Addressed schema compatibility and application testing

2. Backup Automation Script

- Daily backup with timestamp and compression
- Retention policy to remove old backups
- Log monitoring for failures

3. Stored Procedures & Maintenance

- Sample SQL procedures for reporting and cleanup
- Includes documentation for usage and testing

Sample Script Examples

Backup Script (Bash + MySQL)

```
#!/bin/bash
DB_NAME="production_db"
DB_USER="backup_user"
DB_HOST="localhost"
BACKUP_DIR="./backups"
DATE=$(date +%Y-%m-%d_%H-%M-%S)
mkdir -p $BACKUP_DIR
mysqldump -u $DB_USER -h $DB_HOST $DB_NAME | gzip > $BACKUP_DIR/${DB_NAME}_${DATE}.sql.gz
```

Stored Procedure Example (MySQL)

```
DELIMITER //
CREATE PROCEDURE GetMonthlySales(IN target_month INT)
BEGIN
    SELECT customer_id, SUM(amount) as total_sales
    FROM sales
    WHERE MONTH(sale_date) = target_month
    GROUP BY customer_id;
END //
DELIMITER ;
```

Usage Instructions

- Scripts are ready to run in MySQL environments (adjust credentials as needed)
- Backup scripts should be scheduled using `cron` for automation
- Stored procedures can be imported into MySQL databases for testing

Contact

- Email: ianmuriiri25@gmail.com
- LinkedIn: [Your LinkedIn URL]

This repository is ready to serve as a professional showcase for database job applications.